Models with non-constant rates

Variability in evolution, competitive reversal, and speciation as catastrophe

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Contents

Chapter 1

Models with non-constant rates

1.1 Introduction

Every model carried through to a result in the preceding chapters holds its rates fixed. A replicative unit is born at rate λ, dies at rate μ and triggers catastrophe at rate δ, and those three numbers are the same at the end of the process as at the beginning. That assumption is what makes the generating functions close, and it buys a great deal: the roots a and b, the burst-size law, the flooding criterion and the removal budget. It is also the first thing an ecologist will question, because in almost every setting one might want to describe, a rate is not a constant — resources run out, competitors adapt, and success carries its own cost.

This chapter asks what happens when a rate is allowed to move. The models collected here were built at different times and stand at different stages of completion, but they sort by one structural feature: *what moves the rate.* Four mechanisms recur, and where a rate is driven by the process's own state, the behaviour turns on whether the feedback exhausts a quantity or replenishes it.

- **Room that is spent and never returned.** Speciation under logistic growth consumes remaining capacity $K - N(t)$, the integrated rate converges, and the species count has a proper limit law (section 1.2).
- **A critical point dragged after the leader.** Resistance in the Pimentel model acts not on a level but on the state of no preferred direction, and so keeps restoring criticality rather than approaching an equilibrium (section 1.3).
- **A potential restocked from outside.** An accumulating potential, consumed per capita and replenished at a constant inflow, yields a cycle in four movements (section 1.4).
- **A potential the population makes for itself.** Take that same hazard and let the inflow be produced by the population's own area, with the cost charged per *type* rather than per head. The cycle becomes a marginal state; the catastrophe divides rather than destroys; and because its products obey the same law, the cycle branches instead of repeating (section 1.5).

The last two are a matched pair, and the chapter is arranged so that they can be read as one movement. They also mark where the mathematics stops. The first two sections carry closed-form results; the second two carry simulations of models that outrun the machinery of the preceding chapters, and say as exactly as they can what would have to be proved.

Three results carry the finished mathematics, and each is stated here so that the derivations may be read knowing where they go.

- For a pure birth process from one founder with per-capita rate $\beta(t)$, the species count S_t is geometric on $\{1, 2, \dots\}$ with parameter $\mathrm{e}^{-\mathcal{B}(t)}$, where $\mathcal{B}(t) = \int_0^t \beta(u)\,\mathrm{d}u$ (proposition 1.2).

The rate law enters the distribution only through its integral.

- Under the logistic rate that integral converges, so the process has a proper limit law and not merely a finite late-time mean: S_∞ is geometric with mean $(K/N_0)^{\sigma K/\varrho}$ (corollary 1.3), depending on the parameters through exactly those two dimensionless groups.
- In the Pimentel competition model the state at which the chain is equally likely to step up as down is available in closed form ((1.21)), and it is unstable. Two trajectories started from the same state diverge exactly when they are near it.

To these the automaton of section 1.5 adds an observation rather than a theorem, and it is the one most worth taking further: a species is driven onto the locus $X_i = \varphi N_i/c$ at which production and dissipation cancel, and *on* that locus its catastrophe hazard is proportional to its area — which is the chapter on the birth–death–catastrophe process's own δX_t. The chapter's least tractable model may therefore reduce to the chapter's own machinery, and section 1.5.6 says what would have to be shown.

The Yule process, its geometric marginal and the method of characteristics are the mathematical introduction's; the logistic speciation model itself, with its mean, is already a worked example there. The birth–death–catastrophe process, the roots a and b and the burst-size law are the chapter on the birth–death–catastrophe process's and the chapter on its distribution theory's. None of that is re-derived here. What this chapter adds is the distribution under a moving speciation rate, the moving critical point of the Pimentel chain, the automaton and its balance point, and the remaining specifications, stated precisely enough to be taken up where they stop.

§	Model	What moves the rate	Status
1.2	Logistic speciation	remaining capacity $K - N(t)$	closed form
1.3	Pimentel competition	evolving resistance, itself driven by the state	closed-form critical point; simulated
1.3.6	Pimentel+	fitness driven by dominance, reduced by species count	specified
1.4	Rise, run, ruin, rejuvenation	potential V; catastrophe by scarcity X_t/V	simulated
1.5	Multiplicative dissipation	potential V_i made by area, spent per type; fragmentation at $\delta X_i/V_i$	simulated
1.A	Birth–plague	loss channel driven by the coupled component Y_t	specified
1.B	Public good	$\lambda(t) = \lambda_0 + \alpha P(t)$, P made by the population	K^* computed

Table 1.1: The models, and the mechanism by which a rate moves in each. The last column is the shape of the chapter: section 1.2 is its mathematical end, and the rows below it halt at a closed-form fragment or at a simulation. The two below the rule are carried as bare specifications in sections 1.A and 1.B, the birth–plague model in two variants, so the count of distinct specifications is eight.

1.1.1 Notation

Symbols are reused across models but never within one section, and never against a thesis-wide meaning. The table records the assignment for the body of the chapter; the two appendix specifications gloss their own parameters where they stand.

§	Symbol	Meaning
all	λ, μ, δ	birth, death and catastrophe rates, as in the chapter on the birth–death–catastrophe process
	a, b	roots of the characteristic quadratic, $b < 1 < a$ (**??**)
	$\mathcal{K}$	burst size of a birth–death–catastrophe process
1.2	S_t	species count, $S_0 = 1$
	$N(t), K, N_0$	total abundance, carrying capacity, initial abundance
	ϱ	logistic growth rate
	σ	speciation coefficient; $\beta(t) = \sigma(K - N(t))$
	ξ	the constant $K/N_0 - 1$
	$\beta(t)$	speciation rate per species; β is this and nothing else
	$\mathcal{B}(t)$	integrated rate $\int_0^t \beta(u)\,\mathrm{d}u$; η is a constant floor added to β in two of the alternative laws
1.3, 1.3.6	A_t, B_t	counts of the two competing populations, $A_t + B_t = N$
	N	capacity of the enclosure, an integer
	R_a, R_b	resistance of each population
	p_a, p_b	blocking probabilities, at most one non-zero
	ε, α	evolution rate (the step size of the resistance update) and the exponent sharpening that drive near a boundary
	i_c	the critical state (1.21)
	f_a, f_b	fitness, the Pimentel+ recasting of R_a, R_b
	$S_t^{(a)}, S_t^{(b)}$	per-population species counts; ζ couples dominance to speciation and γ penalises fitness per species
1.4	$X_t, V(t)$	count of replicative agencies, and accumulated potential
	ϕ, c, ω	potential inflow, potential consumed per agency, revival rate out of the empty state
1.5	L	side of the lattice; L^2 cells, Moore neighbourhood, periodic
	N_i, X_i, V_i	cells held by species i, its extant sub-species, and its accumulated potential
	φ, c	potential supplied per cell, and spent per extant sub-species; V_0 is the potential a fragment is reset to
	m	mutation coefficient; the per-cell rate is mV_i
	θ, ν	neighbour-support exponent, and the rare-type advantage within a species
	κ_i	mean sub-species area N_i/X_i; d is the resistance of empty ground

Table 1.2: Notation, organised by model. The rates λ, μ and δ, the roots a and b and the burst size $\mathcal{K}$ carry their thesis-wide meanings everywhere in this chapter and are never redefined. V and c are shared between sections 1.4 and 1.5 with the *same* meanings, a potential and the rate at which it is spent, which is the point of the pairing: what differs between the two models is that section 1.5 charges the inflow per cell and the outflow per type. Where one letter genuinely serves two models — α as an exponent in sections 1.3 and 1.3.6, N as an abundance in section 1.2 and an integer capacity in section 1.3, X as a count of individuals in section 1.4 and of types in section 1.5 — the two uses are separated by this table and never occur in the same section.

Section 1.2 takes the moving speciation rate as far as it will go, which is a proper limit law. Section 1.3 obtains the moving critical point of the Pimentel chain and, in section 1.3.6, specifies the coupling that would let excess variation weaken a winner. Sections 1.4 and 1.5 are the matched pair: the first drives catastrophe by a potential restocked from outside, the second by one the population makes for itself, and the second is where the chapter's open questions concentrate. Sections 1.A and 1.B carry two further specifications in bare form, and section 1.C gives the generating-function proof of proposition 1.2.

1.2 Logistic speciation

1.2.1 Variation, and what suppresses it

The premise of the model is Fisher's: variation is in constant supply, and what limits the number of distinct types is not the rate at which they arise but the severity of the conditions under which they must persist [1]. "To maintain a stationary variance, fresh mutations must be available in each generation" is his statement of the first half; the second half is that selection pressure, acting continuously, holds a population near a current optimum and prevents most of that variation from establishing. Nietzsche made the same observation from the breeder's side, that species given superabundant nourishment "tend in the most marked way to develop variation" [2]. Read that way, an abundance of resources is less a cause of diversification than a withdrawal of what had been suppressing it, and the *speciation rate* becomes a function of how much room is left.

That is a claim about the fossil record as much as about breeding. The signature — high early diversification after the opening of ecological space, then a slowing — is the one named by punctuated equilibria [3], and it is the one density-dependent diversification models are built to produce [4]. Whether it is the typical pattern, and at what taxonomic level, is not settled here.

1.2.2 The push of the past

There is an existing account of why diversification rates appear high at the origin of a clade and decline towards the present, and the model below is a response to it rather than a competitor. Mann and Budd, reviewing the phenomenon introduced by Nee and co-workers [5], name it the push of the past [6]:

> The POTPa emerges as a feature of diversification by the fact that all modern clades (tautologically) survived until the present day. This singles them out from the total population of clades that could be generated from any particular pair of background birth and death rates: clades that happened by chance to start off with higher net rates of diversification have better-than-average chances of surviving until the present day. As Nee et al. (1994a) put it, such clades "got off to a flying start"; and they accumulate species faster than one would expect. Once clades become established, they are less vulnerable to random changes in the observed diversification rate, and this value therefore tends to revert to the background rate through time. Long-lived clades thus tend to show high observed rates of diversification at their origin, which then decrease to their long-term average as the present is approached.

The push of the past is a survivorship effect: the rates never move, and the apparent decline is an artefact of conditioning on survival to the present. The model of this section supposes instead that the rate really does move, declining because the ecological room that permitted the early diversification is used up. The two accounts are not exclusive and nothing here argues against the first. What the model offers is a second mechanism with the same signature, and a distribution against which it might one day be told apart from the first [7].

1.2.3 The model

Let $N(t)$ be the total abundance of individuals across all species, taken as a real number on the grounds that it is large compared with the number of species, and let it grow logistically,

$$\frac{\mathrm{d}N}{\mathrm{d}t} = \varrho N\left(1 - \frac{N}{K}\right), \qquad N(0) = N_0, \qquad 0 < N_0 < K, \tag{1.1}$$

with K the carrying capacity of the environment and ϱ the growth rate. Its solution is

$$N(t) = \frac{K\mathrm{e}^{\varrho t}}{\xi + \mathrm{e}^{\varrho t}}, \qquad \xi = \frac{K}{N_0} - 1. \tag{1.2}$$

Let S_t count species, with $S_0 = 1$. Each existing species gives rise to a new one independently at the per-species rate $\beta(t)$, and no species becomes extinct, so the transitions in a vanishing interval are

$$\begin{aligned} \mathbb{P}(S_{t+\Delta t} - S_t = 1 \mid S_t = n) &= \beta(t)\, n\, \Delta t + o(\Delta t), \\ \mathbb{P}(S_{t+\Delta t} - S_t = 0 \mid S_t = n) &= 1 - \beta(t)\, n\, \Delta t + o(\Delta t). \end{aligned} \tag{1.3}$$

This is the Yule process of **??** with its constant rate replaced by a function of time; setting $\beta(t) = \beta$ recovers it exactly. The substitution is not new in itself — taking $\beta(t) = \mathrm{e}^{-t}$ makes the mean of S_t follow Gompertz's 1825 law [8] — and the Yule process was itself introduced in 1925 to model exactly this quantity, the number of species within a genus [9]. What is being tried here is a rate tied to the logistic environment:

$$\beta(t) = \sigma\big(K - N(t)\big), \tag{1.4}$$

which by (1.2) is

$$\beta(t) = \sigma K\, \frac{\xi}{\xi + \mathrm{e}^{\varrho t}}, \tag{1.5}$$

decreasing from $\sigma(K - N_0)$ to zero as the environment fills. The coupling runs one way only: the environment sets the speciation clock, and S_t does not act back on $N(t)$. The same model appears as a worked example in **??**, where its mean is recorded; what is added here is its distribution.

Remark 1.1 (What the rate law is, and is not, claiming). (1.4) treats resource availability as an *index* of selective pressure and not as a mechanism. It says that when there is room, variants persist more often, and it says nothing about how. The model is correspondingly agnostic about the direction of causation: it does not distinguish logistic expansion driving diversification from logistic expansion merely creating the space in which diversification happens.

Underneath that sits a question of timescale. If population expansion is orders of magnitude faster than speciation — which for most plausible parameter values it is — then the logistic phase is over before the species count has moved appreciably, and (1.4) is better read as setting the initial condition for a density-dependent diversification model than as driving the rate through time [4]. Nothing below depends on resolving that, since the mathematics of proposition 1.2 holds for any locally integrable β, and the question is which β the biology warrants.

1.2.4 The marginal law

The first result is that the rate law enters the distribution of S_t only through its integral, and that the resulting law is geometric. Write

$$\mathcal{B}(t) = \int_0^t \beta(u)\, \mathrm{d}u \tag{1.6}$$

for the integrated speciation rate, and assume only that β is locally integrable and non-negative.

Proposition 1.2 (Marginal law of an inhomogeneous Yule process). *Let S_t be a pure birth process on $\{1, 2, \dots\}$ with $S_0 = 1$ and time-dependent per-capita rate $\beta(t)$. Then S_t is geometric on $\{1, 2, \dots\}$ with parameter $\mathrm{e}^{-\mathcal{B}(t)}$:*

$$\mathbb{P}(S_t = n) = \mathrm{e}^{-\mathcal{B}(t)}\big(1 - \mathrm{e}^{-\mathcal{B}(t)}\big)^{n-1}, \qquad n \geq 1, \tag{1.7}$$

with

$$\mathbb{E}(S_t) = \mathrm{e}^{\mathcal{B}(t)}, \qquad \mathrm{Var}(S_t) = \mathrm{e}^{\mathcal{B}(t)}\big(\mathrm{e}^{\mathcal{B}(t)} - 1\big).$$

The proof is the generating-function computation of section 1.C: the forward Kolmogorov equation still yields to the method of characteristics, and the probability generating function is that of a geometric law on $\{1, 2, \dots\}$ with success parameter $\mathrm{e}^{-\mathcal{B}(t)}$. Three things are worth recording here. The first is that the constant-rate Yule law **??** is the case $\mathcal{B}(t) = \beta t$, so nothing is being rebuilt: the substitution $\beta t \mapsto \mathcal{B}(t)$ is exact and not merely a convenience. The second is that the argument nowhere uses the form of β, only that its integral exists, so two rate laws with the same integral to time t give the same distribution at t however differently they arrived there. The third is that the variance falls out of the same line, and had not been recorded before. The standard deviation is $\mathrm{e}^{\mathcal{B}}\sqrt{1 - \mathrm{e}^{-\mathcal{B}}}$, of the same order as the mean, so to fit this model to a single lineage is to fit a mean to one draw from a law whose coefficient of variation is close to one.

1.2.5 The mean, directly

With the proposition in hand $\mathbb{E}(S_t) = \mathrm{e}^{\mathcal{B}(t)}$ is a corollary, and it is also **??** with $\mu \equiv 0$. The direct derivation is retained as a check on the integral that follows.

Over a vanishing interval the mean after is the mean before plus the expected change, and by (1.3) the expected change is $\beta(t)\mathbb{E}(S_t)\,\Delta t + o(\Delta t)$. Dividing by Δt and letting it vanish,

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}(S_t) = \beta(t)\mathbb{E}(S_t)\,, \tag{1.8}$$

which separates. With $\mathbb{E}(S_0) = 1$ the solution is $\mathbb{E}(S_t) = \mathrm{e}^{\mathcal{B}(t)}$, and for the logistic rate law (1.5) the integral is elementary:

$$\int \frac{\mathrm{d}t}{\xi + \mathrm{e}^{\varrho t}} = \frac{1}{\xi}\left(t - \frac{1}{\varrho}\ln\left(\xi + \mathrm{e}^{\varrho t}\right)\right), \tag{1.9}$$

so that

$$\mathcal{B}(t) = \sigma K\left(t - \frac{1}{\varrho}\ln\left(\xi + \mathrm{e}^{\varrho t}\right) + \frac{1}{\varrho}\ln\left(\xi + 1\right)\right), \qquad \mathbb{E}(S_t) = \mathrm{e}^{\sigma K t}\left(\frac{\xi + 1}{\xi + \mathrm{e}^{\varrho t}}\right)^{\sigma K/\varrho}. \tag{1.10}$$

1.2.6 The limit law

The question the model was built to answer is what happens at late time, when β has fallen to nothing and the species count has stopped moving. Since S_t is non-decreasing and its mean is bounded, it converges almost surely to a finite limit; the content of the following is that the limit is proper and its law is known.

Corollary 1.3 (Limit law under the logistic rate)**.** *Under (1.4), $\mathcal{B}(t)$ converges as $t \to \infty$, to*

$$\mathcal{B}(\infty) = \frac{\sigma K}{\varrho}\ln\left(\frac{K}{N_0}\right), \tag{1.11}$$

and S_∞ is geometric on $\{1, 2, \dots\}$ with

$$\mathbb{E}(S_\infty) = \left(\frac{K}{N_0}\right)^{\sigma K/\varrho}. \tag{1.12}$$

Proof. From (1.10),

$$t - \frac{1}{\varrho}\ln\left(\xi + \mathrm{e}^{\varrho t}\right) = -\frac{1}{\varrho}\ln\left(1 + \xi\mathrm{e}^{-\varrho t}\right) \longrightarrow 0,$$

so $\mathcal{B}(t) \to (\sigma K/\varrho)\ln(\xi + 1)$, and $\xi + 1 = K/N_0$. The law is (1.7) with $\mathcal{B}(t)$ replaced by its limit, which is legitimate because S_t increases to S_∞ and the geometric probabilities are continuous in the parameter. □

The dependence on the parameters is now explicit, and it is simpler than the five constants $\sigma, K, \varrho, N_0, \xi$ suggest: only two dimensionless groups survive. The first is K/N_0, the room the genus had at the outset, so that a clade founded into a nearly empty environment integrates more unused capacity than one founded into a full one and ends with more species. The second is $\sigma K/\varrho$, the speciation capacity per unit of population growth rate, which sets how much of that room is converted before it closes; doubling the growth rate and the speciation coefficient together changes nothing. That is the answer to the question posed but not settled when the model was written — how the asymptotic behaviour depends on the parameters and their interplay — and it is an answer about the whole distribution rather than about the mean alone.

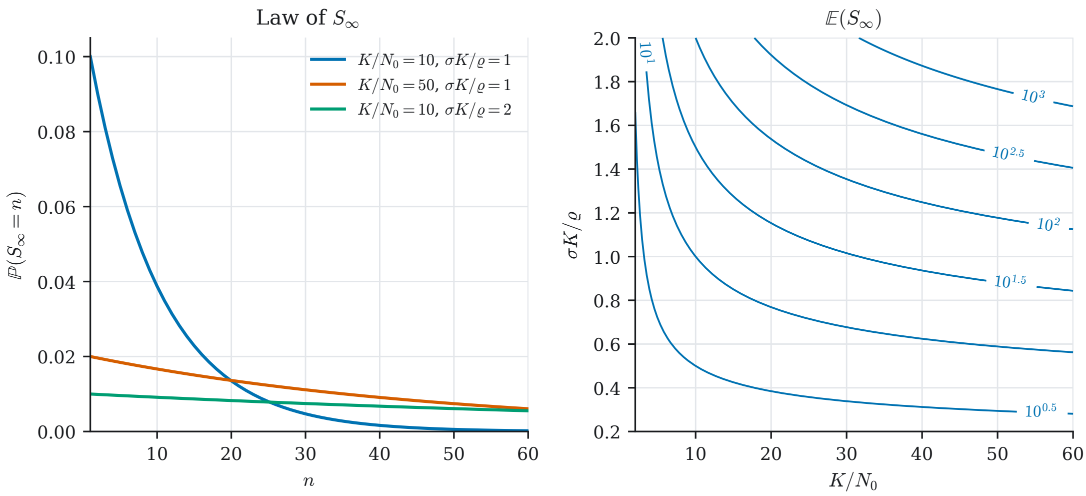


Figure 1.1: What corollary 1.3 says, drawn from the closed form. Panel (a): the law of S_∞ is geometric, so most of its mass sits at small counts however large the mean, and its coefficient of variation is close to one — fitting this model to a single clade is fitting a mean to one draw from that. Panel (b): the expected final count $\mathbb{E}(S_\infty) = (K/N_0)^{\sigma K/\varrho}$ depends on the five constants $\sigma, K, \varrho, N_0, \xi$ only through the two dimensionless groups on the axes, so every point on a contour is a genus with the same expected diversity reached from different parameters.

1.2.7 The other rate laws

The form (1.4) was one of five candidates written down when the model was set up. The others were

$$\beta_2(t) = \sigma\big(K - N(t)\big) + \eta, \qquad \beta_3(t) = \sigma\,\frac{\mathrm{d}N}{\mathrm{d}t},$$
$$\beta_4(t) = \sigma\,\frac{\mathrm{d}N}{\mathrm{d}t} + \eta, \qquad \beta_5(t) = \sigma\big(K_t - N(t)\big), \tag{1.13}$$

with η a constant floor and K_t a capacity switching at random. The motivation for β_3 and β_4 is that a mutation event needs a replication event, so speciation might track the rate of population increase rather than the room remaining; the floor η in β_2 and β_4 is a baseline propensity to vary that survives the environment filling up. Proposition 1.2 settles two of the four immediately, because it asks only for an integral.

For β_3 the integral is the total change in abundance:

$$\mathcal{B}(\infty) = \sigma \int_0^\infty N'(u)\,\mathrm{d}u = \sigma\,(K - N_0), \tag{1.14}$$

so S_∞ is again geometric, with mean $\mathrm{e}^{\sigma(K-N_0)}$. The comparison with (1.12) is worth a moment. Under β_1 the limiting mean is a power of K/N_0 and under β_3 an exponential of $K - N_0$, so the

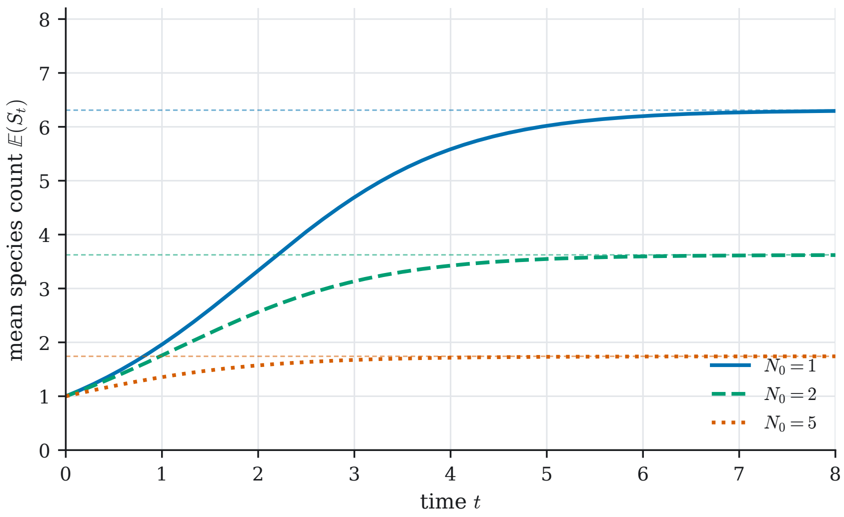


Figure 1.2: Mean species count (1.10) for $N_0 = 1, 2, 5$, with $K = 10$, $\varrho = 1$ and $\sigma = 0.08$; dashed lines are the limits $(K/N_0)^{\sigma K/\varrho}$. A smaller founding abundance leaves more integrated unused capacity, so the limiting mean is higher. This is the pure-birth model of (1.3), not the birth–death numerical variant recorded at **??**; section 1.2.8 is what a constant extinction rate does to this one.

first is sensitive to the *ratio* of capacity to founding abundance and the second to their *difference*; a model fitted under one and read under the other would be badly wrong for large K.

For β_2 and β_4 the floor makes $\mathcal{B}(t) \sim \eta t$, which diverges, and there is then no limit law: S_t remains geometric at every finite time with parameter $\mathrm{e}^{-\mathcal{B}(t)} \to 0$, and the species count grows without bound, eventually at the constant-rate Yule rate η. That is a statement about the model, not a defect of the analysis: a floor on the speciation rate says diversification never stops, and a process that never stops has no final species count for a distribution to be about.

β_5 is the one that remains genuinely open, and the proposition says why. If K_t switches at random then $\mathcal{B}(t)$ is itself a random variable, and (1.7) holds conditionally on it. Unconditionally S_t is a *mixture* of geometric laws over the law of $\mathcal{B}(t)$,

$$\mathbb{P}(S_t = n) = \mathbb{E}\Big(\mathrm{e}^{-\mathcal{B}(t)}\big(1 - \mathrm{e}^{-\mathcal{B}(t)}\big)^{n-1}\Big), \tag{1.15}$$

so that the problem becomes finding the law of the integrated rate under the switching regime. For a finite-state Markov switching K_t that is a Markov-modulated integral and is tractable; for anything else it is not obviously so. Of five candidate rate laws, two are now closed, two are explained non-results, and one is a well-posed problem.

The pattern across all five is the same one: nothing about a moving speciation rate is difficult so long as its integral is deterministic, and everything about it is difficult once the integral is not.

1.2.8 What extinction does

Everything above depends on the process being pure birth. (1.3) has no downward channel, and both the geometric marginal of proposition 1.2 and the limit law of corollary 1.3 come of the fact that $\mathcal{B}$ converges and nothing is ever taken away. It is worth seeing what the same origination law does once species are allowed to go extinct, because the answer is not a slower version of the same thing.

Let each species become extinct independently at the constant per-capita rate μ, so that S_t is a linear birth–death process with a moving birth rate and a fixed death rate. Its mean is the exponential of the net integrated rate,

$$\mathbb{E}(S_t) = \exp\left(\mathcal{B}(t) - \mu t\right), \tag{1.16}$$

which follows from $\mathrm{d}\mathbb{E}(S_t)/\mathrm{d}t = (\beta(t) - \mu)\mathbb{E}(S_t)$ and needs no generating function. Two things follow at once. The mean turns over exactly where the two rates cross, at the t solving $\beta(t) = \mu$, whatever the rate law; and since $\mathcal{B}(\infty)$ is finite under every law of section 1.2.7 while μt is not, $\mathbb{E}(S_t) \to 0$. A constant extinction rate set against an origination rate that runs out takes the clade with it, and there is no limit law of any kind.

Figure 1.3 draws that under $\beta_3(t) = \sigma N'(t)$, with one modification: the extinction of a genus's last species is suppressed, so that the picture is about the shape of the rise and fall rather than about the disappearance of the clade. The peak is unaffected, sitting at the crossing as (1.16) requires, and the mean settles at one species rather than at zero.

What the figure shows is the signature section 1.2.1 began with, and the pure-birth model of section 1.2.3 does not produce it. There the species count rises and saturates, because the rate runs out and nothing erodes what it built; corollary 1.3 is exactly that statement. Here it rises, turns over and decays, which is the shape the fossil record is usually described as having. The two accounts of section 1.2.2 are worth holding against this. The push of the past produces a declining *apparent* rate by conditioning on survival while the true rates stay fixed; (1.4) produces one by moving the rate and keeping every clade. Figure 1.3 is a third thing again — rates that move *and* a real loss — and it is the only one of the three in which the count itself goes down.

Open questions.

1. **The mixture at β_5.** What is the law of $\mathcal{B}(t)$ when K_t switches between finitely many values at exponential rates, and what does the resulting mixture (1.15) do to the tail of S_t? This is the question in this section most worth taking up, and it asks for a Markov-modulated integral rather than for a new model.

2. **Identifiability.** Since only $\mathcal{B}(t)$ enters (1.7), no observation of S_t at a single time can separate σ, K and ϱ beyond the two groups of (1.12). What a time series of S_t at several times does identify, and whether β_1 and β_3 can be told apart from one, is not known.

3. **The timescale question of remark 1.1.** Whether population expansion and speciation are separated by orders of magnitude in any clade for which both are measurable, and therefore whether the model describes a rate or an initial condition, is an empirical question this chapter cannot settle.

4. **The marginal law under extinction.** (1.16) gives the mean but says nothing about the distribution, and proposition 1.2 does not survive a downward channel as it stands. Whether the geometric marginal survives with an atom at zero — so that proposition 1.2 is the $\mu = 0$ case of something more general, and the rate law still enters only through its integral — is the question figure 1.3 raises and does not answer.

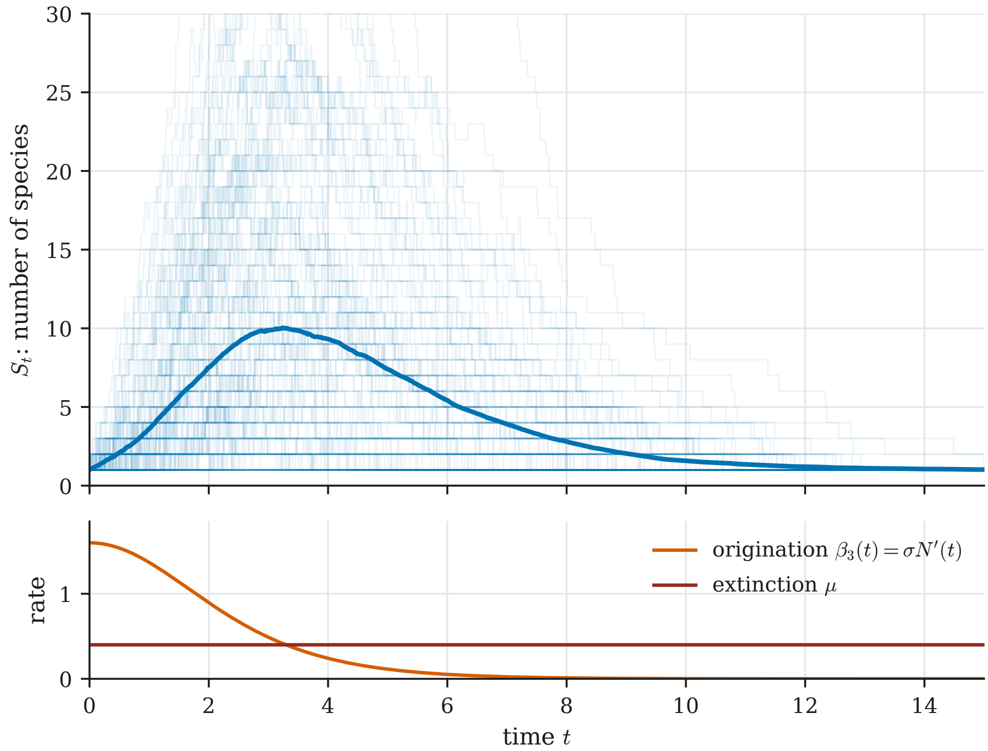


Figure 1.3: Logistic speciation with a constant extinction rate, at $(K, N_0, \varrho, \sigma, \mu) = (10, 5, 0.8, 0.8, 0.4)$ from $S_0 = 1$: 160 exact realisations in pale blue, their mean over 400 paths in bold. Origination follows $\beta_3(t) = \sigma N'(t)$, drawn below against the constant μ. $N_0 = K/2$ starts the abundance at the inflection of (1.1), so origination is at its largest at $t = 0$ and decays from there. The mean peaks at 10.0 species at $t = 3.24$, and the rates cross at $t = 3.29$: the two coincide, as (1.16) requires. Extinction of a genus's last species is suppressed, which is why the paths have a floor at one and the mean settles at 1.02 rather than decaying to zero. Set against the same origination law without extinction, where (1.14) gives a limiting mean of $e^{\sigma(K-N_0)} \approx 55$ approached from below and never left, the cost of a constant μ is the whole limit law.

1.3 Competitive reversal: the Pimentel model

The speciation model of the previous section lets variation out when there is room. This one asks the complement: what happens when two populations share the room, and the minority is the one under selection.

1.3.1 The experiment

Pimentel and colleagues devised and ran an experiment in which two populations of flies are pitted against one another in a fight for domination and, ultimately, survival [10, 11]. Place a colony of houseflies (*Musca domestica*) in one side of a boxed enclosure and a competing number of blowflies (*Phaenicia*) at the other, and give them an ample but finite supply of food. Before long one of the species dominates the niche. The arrangement is unstable, and either the housefly or the blowfly population eventually sees off the other.

Pimentel and his colleagues did something clever to augment the dynamics. They contrived an enclosure — the original box, but with a myriad of internal sub-compartments, a three-dimensional maze of tunnels and small chambers. That arrangement gives the enclosure a desirable property: with the populations initially placed at opposing sides, communication by contact is delayed by a period greater than the average lifespan of either species. In the months taken for a species to spread from one side to the other, multiple generations ensue.

Why is that desirable? By slowing the potential spread, Pimentel imposes the constraint that a total domination, if one is arrived at, cannot be won in one or a few fly lifespans. The war must be won over succeeding generations, grandchildren's grandchildren all fighting for the same cause, and that is what allows the hand of Darwinian selection to come into play.

The observed dynamic was the following. Blowflies were pushed almost to extinction by the initially superior numbers of *Musca domestica*, and then the tide turned: from the brink, the blowfly underdogs clawed back their footing and eventually overthrew the advances of their adversaries. The hypothesis is that during this squeeze of reduced population the blowflies began to select, among their limited numbers, for the abilities favouring direct competition with their adversaries. The houseflies, doing well as they were, were subject to no such channelling selection; they became complacent while their foes grew strong, cut into shape by a harshly oppressive environment.

It has not been achieved experimentally, but Pimentel proposed that with the enclosure arranged correctly an oscillation could be set up, in which the upper hand passes continually from one side to the other, each being pushed to the brink and moulded to competitive advantage in times of scarcity — a continual arms race for survival. The question this section asks is whether a Markov chain with evolving blocking probabilities can reverse in the same way.

1.3.2 The model

Consider two populations competing for space in a finite domain they both occupy. Let A_t count one of them, labelled a, and B_t the other, labelled b, both in discrete time $t \in \mathbb{N} \cup \{0\}$. Finiteness of the enclosure is imposed by $A_t + B_t \leq N$ for a maximum capacity N, and with the initial condition $A_0 = B_0 = N/2$ for even N the two counts are tied by $B_t = N - A_t$ throughout, so it is enough to follow A_t. Its transition probabilities are

$$\mathbb{P}(A_{t+1} = j \mid A_t = i) = \begin{cases} q_a = \dfrac{i}{N}(1 - p_b), & j = i + 1, \\ q_b = \left(1 - \dfrac{i}{N}\right)(1 - p_a), & j = i - 1, \\ q_c = \dfrac{i}{N} p_b + \left(1 - \dfrac{i}{N}\right) p_a, & j = i, \\ 0 & \text{otherwise}, \end{cases} \tag{1.17}$$

for $0 < i < N$, with $i = 0$ and $i = N$ absorbing.

The three cases come from the following step rule, viewed from the perspective of population a.

Whose move. With probability i/N population a attempts to advance; with probability $1 - i/N$ the opponent attempts to, potentially forcing a back one.

Advancing. If a has decided to advance it does so, unless it is blocked, which happens with probability p_b.

Being advanced upon. If the opponent is advancing, a blocks with probability p_a; if it fails to, it is forced back a step.

Stalemate. If a block is effective, both parties remain where they are, and $A_{t+1} = A_t$.

The capacity to block is the model's evolving quantity. Let it depend on a *resistance* R_k for $k = a, b$ through

$$p_a = \max\left\{0,\, 1 - \frac{R_b}{R_a}\right\}, \qquad p_b = \max\left\{0,\, 1 - \frac{R_a}{R_b}\right\}, \tag{1.18}$$

so that at most one of the two is non-zero: whichever population has the higher resistance blocks, and the other does not. If $R_a \gg R_b$ then p_a is close to one and p_b is zero.

Resistances are driven up when the associated population is in a minority. They are updated at each step by

$$R_{k,t+1} = R_{k,t} + \varepsilon\,\Delta R_{k,t}, \tag{1.19}$$

at an evolution rate ε, with

$$\Delta R_{a,t} = \max\left\{0, \left(\frac{N - A_t}{A_t}\right)^{\alpha}\right\}, \qquad \Delta R_{b,t} = \max\left\{0, \left(\frac{A_t}{N - A_t}\right)^{\alpha}\right\}. \tag{1.20}$$

Each increment grows as its population is squeezed towards its absorbing boundary, and is larger for the population in the minority. Only the ratio R_b/R_a enters (1.18), so what drives the blocking probabilities is the *difference* between the two increments rather than either alone. The mechanism intended is the one Pimentel proposed: most of the competitive interactions experienced by members of the smaller population are with the opposition, and the drive for selection towards competitive ability is correspondingly stronger, while the majority population meets mostly its own kind and is under no such pressure. The exponent α sharpens that drive near the boundaries. Taking $\alpha = 1$ makes the increment simply proportional to the ratio of interspecific to intraspecific encounters, which is the physical case; $\alpha > 1$ is retained because it is what makes the criticality of section 1.3.4 visible.

The initial resistances need saying, because the model is not scale-free in them. Since (1.18) depends on R_a and R_b only through their ratio, while (1.19) adds an absolute increment, scaling $R_{a,0}$, $R_{b,0}$ and ε by a common factor leaves every trajectory unchanged: the two enter only through ε/R_0. Taking $R_{a,0} = R_{b,0} = 1$ throughout therefore costs nothing and fixes what ε means.

Two things about (1.17), (1.18) and (1.20) are worth naming, since they are what make this a chapter about moving rates. The first is that the step probabilities are functions of the resistances and the resistances are functions of the state, so the transition kernel at time t depends on the whole path up to t and not on A_t alone. The second follows: the triple $(A_t, R_{a,t}, R_{b,t})$ is Markov, and A_t by itself is not.

1.3.3 Results

Simulated at $\alpha = 1$, the model does what Pimentel observed. Both realisations in figure 1.4 take the same shape: one population reaches a position of real advantage, and does not keep it. Trained by interspecific competition, the oppressed population makes a striking comeback and overthrows its adversary completely. At this exponent the reversal happens once or twice and then the chain is absorbed, so what Pimentel hoped for arrives as an episode rather than as a cycle.

Raising α sharpens the resistance drive near the boundaries, and with it the sensitivity of a trajectory to where it starts. Figure 1.5 makes that comparison at $\alpha = 10$ and $N = 10^5$, running two independent realisations from each of two initial conditions: one with the populations evenly matched, one with a bias imposed at the outset. Started well matched, the two realisations diverge substantially; started biased, they stay close for far longer. What differs is not the noise, which is the same in both, but where the trajectory sits relative to the state at which the chain has no preferred direction.

Raising α also buys far more of the behaviour the model was built for. Where $\alpha = 1$ gives one reversal or two, $\alpha = 10$ buys of order ten reversals before fixation, and the swing grows rather than decays. Figure 1.5 under-shows that run: the full excursion, at the same parameters, continues for several million further steps before absorption. The oscillation Pimentel conjectured is therefore not absent from the model, only impermanent. Every run here still fixates in the end, and what raising α moves is how long the alternation lasts before it does.

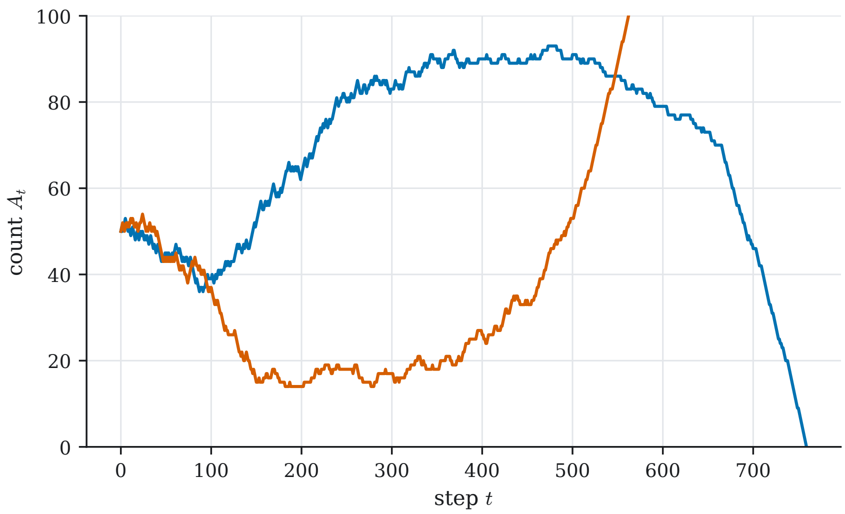


Figure 1.4: Two realisations of the Pimentel chain at $\alpha = 1$, $N = 100$, $\varepsilon = 10$, the physical case. Each reverses: an early advantage is not kept, and the two histories, started from the same symmetric state, absorb at opposite boundaries.

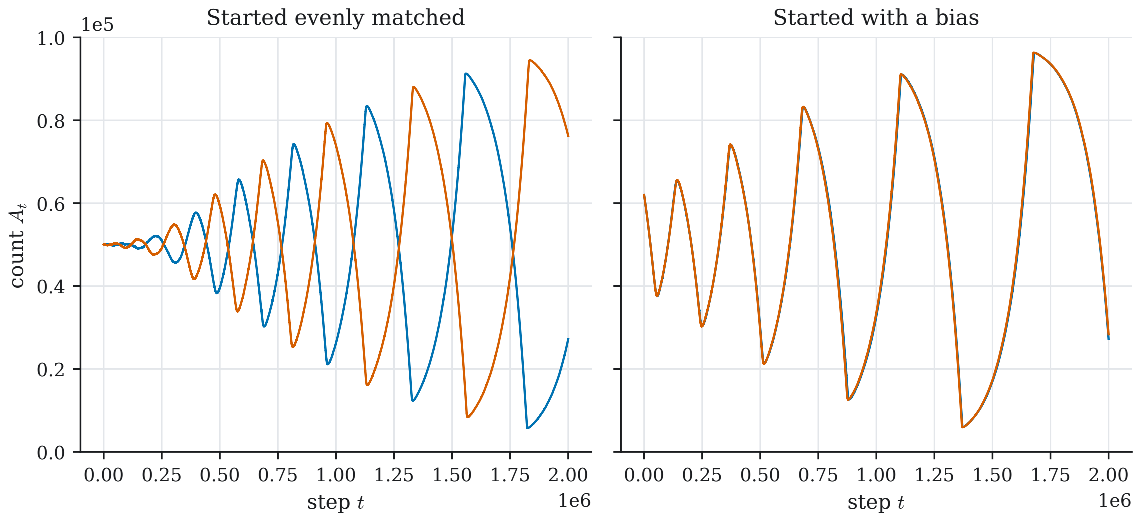


Figure 1.5: Two independent realisations from each of two initial conditions, at $\alpha = 10$, $N = 10^5$, $\varepsilon = 1$, over 2×10^6 steps. Panel (a): started evenly matched, and so near the critical state, the runs separate early and stay apart. Panel (b): started with one in the majority, and so away from it, they track one another. Proximity to the critical state, not the size of the noise, decides how far apart two histories of the same model end up.

1.3.4 The moving critical point

That state can be located exactly. The chain is at its *critical point* when a step up and a step down are equally likely, that is when $q_a = q_b$ in (1.17). Writing i_c for the state at which this holds and using the mutual exclusivity of p_a and p_b,

$$i_c = \begin{cases} N\dfrac{1-p_a}{2-p_a}, & p_a > 0 \text{ and } p_b = 0, \\ N\dfrac{1}{2-p_b}, & p_b > 0 \text{ and } p_a = 0. \end{cases} \tag{1.21}$$

When neither population blocks, $p_a = p_b = 0$ and $i_c = N/2$, the symmetric point. Blocking moves it off centre, and moves it towards the boundary the *leading* population is heading for: if a is in the majority then it is b that accumulates resistance, so $p_b > 0$ and $i_c = N/(2 - p_b)$ climbs towards N. And i_c is an *unstable* point — comparing q_a with q_b in (1.17) gives $q_a > q_b$ exactly when $i > i_c$, in both cases of (1.21) — so the chain is carried away from it rather than returned to it. A reversal is what happens when i_c, climbing after the leader, overtakes the trajectory, and the drift that was carrying the majority forward begins to carry it back.

(1.21) is less analytic than it looks. Since p_a and p_b depend on R_a and R_b, and those are accumulated along a stochastic trajectory by (1.19), i_c is a function of the path and not of the parameters alone. What it is not is intractable. Given a realisation, the sequence $i_{c,t}$ is computed step by step at the cost of the simulation itself and can then be drawn over the trajectory that produced it, which is what figure 1.6 does.

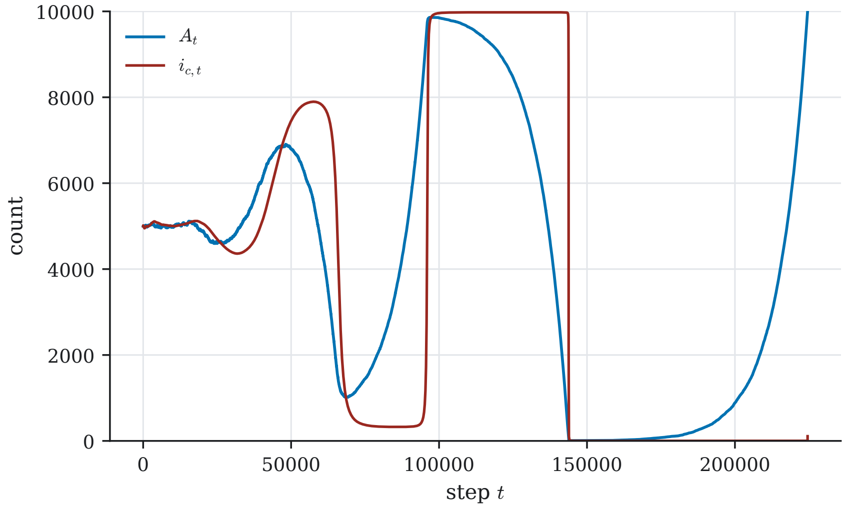


Figure 1.6: A single realisation at $\alpha = 3$, $N = 10^4$, $\varepsilon = 1$, with the critical state $i_{c,t}$ of (1.21) overlaid. The critical state does not stay put. It is dragged towards whichever boundary the leading population is heading for, and saturates near 0 or near N while that population is well ahead; each reversal of A_t follows $i_{c,t}$ crossing it.

Read against figure 1.5, figure 1.6 makes the mechanism plain: trajectories diverge exactly where A_t is near $i_{c,t}$, because there the drift that would otherwise restore them is absent. That is what the resistance dynamics buy. An ordinary random walk on $\{0, \ldots, N\}$ is critical only if its step probabilities happen to be balanced, whereas here the resistance update *drags the critical point after the leading population* and so keeps returning the process to a state at which its own future is undetermined. The reversal Pimentel observed is therefore not a deterministic consequence of the squeeze; it is what a squeeze does to a chain that the squeeze itself has made critical.

1.3.5 A spatial version

The model above has no geometry, where Pimentel's enclosure had a great deal of it. A spatial version was therefore written as a cellular automaton, in `C++` using the SFML graphics library, each cell holding one of the two populations and resistances being inherited and updated locally by the same rule; figure 1.7 shows six frames from it. A model of evolutionary reversal in competitive dominance had been proposed before [12]; the model here was not derived from it and differs in form.

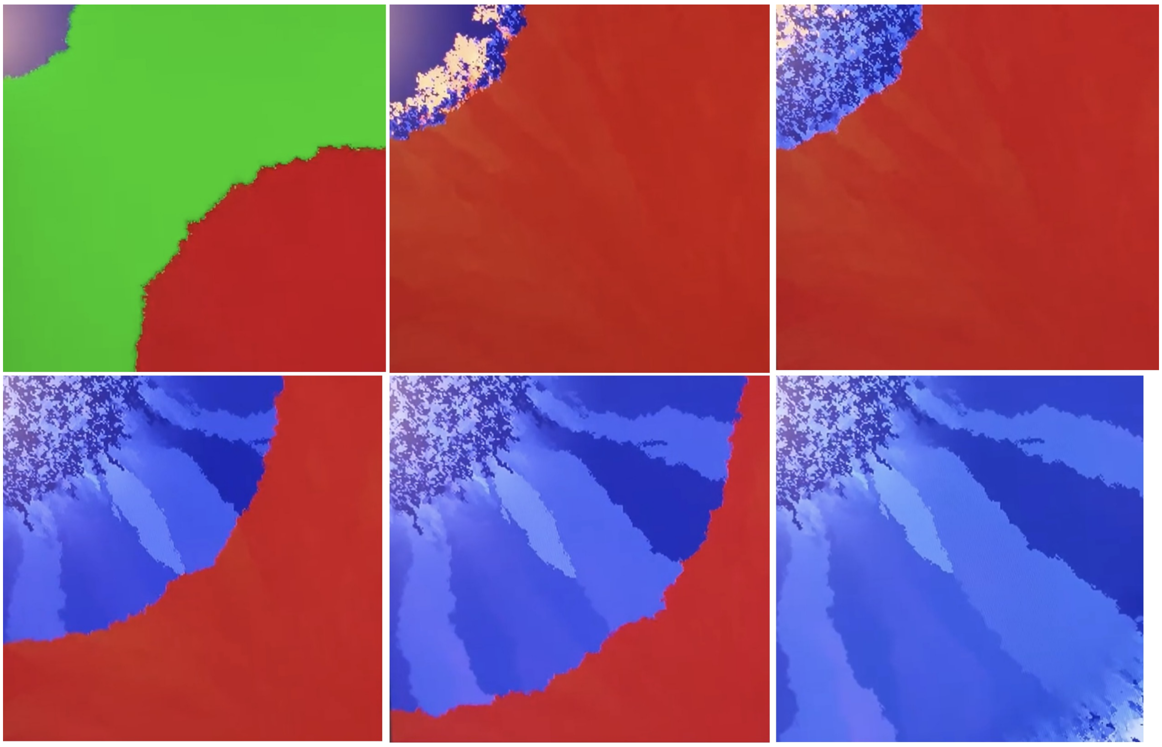


Figure 1.7: Six frames from the spatial version, left to right along the top row then the bottom: advance, squeeze, fringe, crossing, recovery, fixation. Unoccupied cells are green in the first frame. Red advances until blue is confined to one corner, where a pale fringe of elevated resistance appears; blue then crosses back and takes the domain. The reversal is spatial as well as numerical.

1.3.6 Coupling speciation to competition

Section 1.3 makes a losing population strong, the squeeze being what drives its resistance up. This subsection asks the complement, whether a winning population can be made weak by a mechanism of the same kind; the hypothesis it states is that excess variation weakens an established type, so that a population which is winning diversifies and thereby loses. The claim is Nietzsche's, put into a model: a type is made hard by long struggle against constant unfavourable conditions, favourable conditions relax the constraint which made it hard, and the resulting exuberance of variation is itself the mechanism of decline [2]. What is worth retaining is the shape of the claim: that the decline is *endogenous*, a consequence of the success preceding it rather than a response to any external shock.

The model is the kernel of section 1.3.2, recast so that the quantity under selection is fitness rather than resistance, with a species count carried alongside and allowed to act back on that fitness. Written as step probabilities for A_t,

$$\mathbb{P}(\Delta A_t = 1) = \frac{A_t}{N}\,\frac{f_a}{\bar{f}}, \qquad \mathbb{P}(\Delta A_t = -1) = \left(1 - \frac{A_t}{N}\right)\frac{f_b}{\bar{f}}, \tag{1.22}$$

with $\bar{f}$ the mean fitness across the domain and, under the Moran assumption, $A_t = N - B_t$. Each probability is again the chance of meeting a member of the opposing group multiplied by the chance of prevailing against them, as in (1.17); what has changed is that prevailing is now governed by a fitness ratio rather than by a blocking probability. Fitness is driven up by minority status exactly as resistance was:

$$f_a(t+\Delta t) = f_a(t) + \varepsilon\left(\frac{B_t}{A_t}\right)^{\alpha}, \qquad f_b(t+\Delta t) = f_b(t) + \varepsilon\left(\frac{A_t}{B_t}\right)^{\alpha}. \tag{1.23}$$

Fitness here is the same evolving quantity as the resistance R_a, R_b of the previous section, under a second name: the update rule and its exponent are identical, and the increment coefficient is ε in both. The two kernels are not literally identical, since (1.17) has a stalemate outcome and (1.22) has none; both are carried because the fitness form is the one that admits the coupling below, in which a scalar can be added to and subtracted from.

Now give each population its own species count, $S_t^{(a)}$ and $S_t^{(b)}$, evolving as birth–death processes with constant per-species extinction μ_a and a speciation rate that rises with both dominance and fitness:

$$\mathbb{P}\Big(\Delta S_t^{(a)} = 1\Big) = \beta_a S_t^{(a)}\,\Delta t, \qquad \mathbb{P}\Big(\Delta S_t^{(a)} = -1\Big) = \mu_a S_t^{(a)}\,\Delta t, \qquad \beta_a = \left(\frac{A_t}{N}\right)^{\zeta} f_a, \tag{1.24}$$

and correspondingly for b with A_t replaced by B_t, the exponent ζ controlling how firmly dominance must be established before variation is released. This is the premise of section 1.2 transplanted into a competitive setting: variation is in constant supply, and what releases it is the easing of whatever was suppressing it, measured here by the share of the domain a population controls.

The coupling that closes the loop is a penalty on fitness proportional to the species count:

$$f_a(t+\Delta t) = f_a(t) + \varepsilon\left(\frac{B_t}{A_t}\right)^{\alpha} - \gamma S_t^{(a)}, \qquad f_b(t+\Delta t) = f_b(t) + \varepsilon\left(\frac{A_t}{B_t}\right)^{\alpha} - \gamma S_t^{(b)}, \tag{1.25}$$

with γ the amount by which the inefficiency of carrying excess variation weighs on competitive performance.

That is the whole specification, and it has never been simulated. Both routes to reversal are present in it at once: the squeeze that strengthens the loser, through (1.23), and the variation that weakens the winner, through the penalty of (1.25). Both favour whichever population is behind, so the model does not say in advance which of them does the work, and that is what makes it worth running.

Open questions.

1. **Whether collapse by variation happens at all.** Does the $-\gamma S_t^{(a)}$ term in (1.25) ever produce a reversal that the squeeze-driven mechanism of section 1.3 would not have produced unaided, or does it only slow that one down? Separating the two means holding everything else fixed and comparing runs at $\gamma = 0$ against runs at $\gamma > 0$, which asks for simulation and for nothing else.

2. **The relevant regime.** Collapse by variation should require $\gamma S_t^{(a)}$ to grow comparable with the accumulated fitness advantage over the timescale on which $S_t^{(a)}$ itself grows, which ties γ, ζ and μ_a together. Whether that regime is a thin sliver of parameter space or a broad band of it follows directly from the first question.

Open questions.

1. **Fixation.** The fixation probability and the expected time to fixation, as functions of (α, ε, N), were never computed. The first-step machinery of **??** applies to the enlarged chain $(A_t, R_{a,t}, R_{b,t})$, which is where the work would start.

2. **Whether the oscillation persists.** Every realisation run here eventually fixated, at $\alpha = 1$ after an episode and at $\alpha = 10$ after of order ten reversals. So the question is not whether the alternation exists but whether it can be made to last: does the expected number of reversals before fixation grow without bound in α, ε or N, or does it saturate? This is the sharpest question in the section, because it is the one the model was built to answer.

3. **A diffusion limit.** For large N there should be a diffusion approximation in which (1.21) becomes a moving unstable equilibrium and the divergence of nearby trajectories is a statement about a local Lyapunov exponent. That would turn the qualitative observation of figure 1.6 into a rate.

1.4 Rise, run, ruin, rejuvenation

1.4.1 The cycle the model is for

Where the two competition models work inside an arena of fixed capacity, this one replaces the capacity with a flow, and the flow buys a cycle in four movements. *Potential* accumulates while it goes unused; replicative agencies proliferate into the surplus once there is one; their proliferation consumes the potential until scarcity brings a catastrophe in which the agencies are lost; and the potential rebuilds until a revival restarts the wave.

The idea is Prigogine's. In a system through which energy or matter flows, structures can emerge that appear to run against the second law: order appears locally, as a spontaneous reduction of entropy, sustained by the throughput and destroyed if the throughput stops [13]. Such *dissipative structures* are characterised by chaotic dynamics, in which small fluctuations are amplified into the details of the eventual form. Whether the fragmentation of a genus into species is an instance of one has been argued [14, 15], and the model below does not settle it; the claim made here is the weaker one, that the four-part cycle is a recognisable shape worth writing down as a Markov chain. The pair $(X_t, V(t))$ is a birth–death–catastrophe process with catastrophe rate $\delta X_t/V(t)$ and an auxiliary ODE for V, rather than a new class of object.

1.4.2 The model

Let X_t count replicative agencies and $V(t)$ the accumulated potential. Potential flows in at a constant rate ϕ and is consumed at a rate c per agency:

$$\frac{\mathrm{d}V}{\mathrm{d}t} = \phi - c\,X_t. \tag{1.26}$$

Given V, the count X_t moves by four channels, of which the second is the one that carries the model:

$$\begin{aligned}
\mathbb{P}(\Delta X_t = 1 \mid X_t) &= \lambda X_t\,\Delta t + o(\Delta t) && \text{proliferation,}\\
\mathbb{P}(X_{t+\Delta t} = 0 \mid X_t) &= \delta\,\frac{X_t}{V(t)}\,\Delta t + o(\Delta t) && \text{catastrophe,}\\
\mathbb{P}(X_{t+\Delta t} = 1 \mid X_t = 0) &= \omega\,\Delta t + o(\Delta t) && \text{revival,}\\
\mathbb{P}(X_{t+\Delta t} = X_t) &= 1 - \left(\lambda X_t + \delta\frac{X_t}{V(t)} + \omega\mathbf{1}\{X_t = 0\}\right)\Delta t + o(\Delta t) && \text{no event.}
\end{aligned} \tag{1.27}$$

The catastrophe channel is the model's whole content. Its rate is not δX_t, as it would be in the birth–death–catastrophe process of the chapter on the birth–death–catastrophe process, but $\delta X_t/V(t)$, so that catastrophe is driven by per-capita scarcity: a large population sitting on a large potential is no more at risk than a small one on a small potential, while a population that has consumed its potential is at risk in proportion to how thoroughly it has done so. That is what makes the cycle close rather than run once. Revival at rate ω out of the empty state keeps 0 from being absorbing, and is what makes the process recurrent.

The potential cannot in fact be exhausted, which is what makes the model well posed. As $V \downarrow 0$ with $X_t \geq 1$ the catastrophe hazard $\delta X_t/V(t)$ behaves like $\delta/\big(c(t^* - t)\big)$, where t^* is the time (1.26) would reach zero; that integrates to infinity, so catastrophe arrives almost surely first and V stays positive for all time.

Neither nonlinearity here is available to the machinery of the preceding chapters. The catastrophe rate depends on V, and V depends on the whole history of X through (1.26), so that the pair $(X_t, V(t))$ is Markov while X_t alone is not and no generating function for X_t closes.

1.4.3 What the simulations show

Figures 1.8 and 1.9 are exact realisations. They show the cycle running, and they show that its period is not a constant.

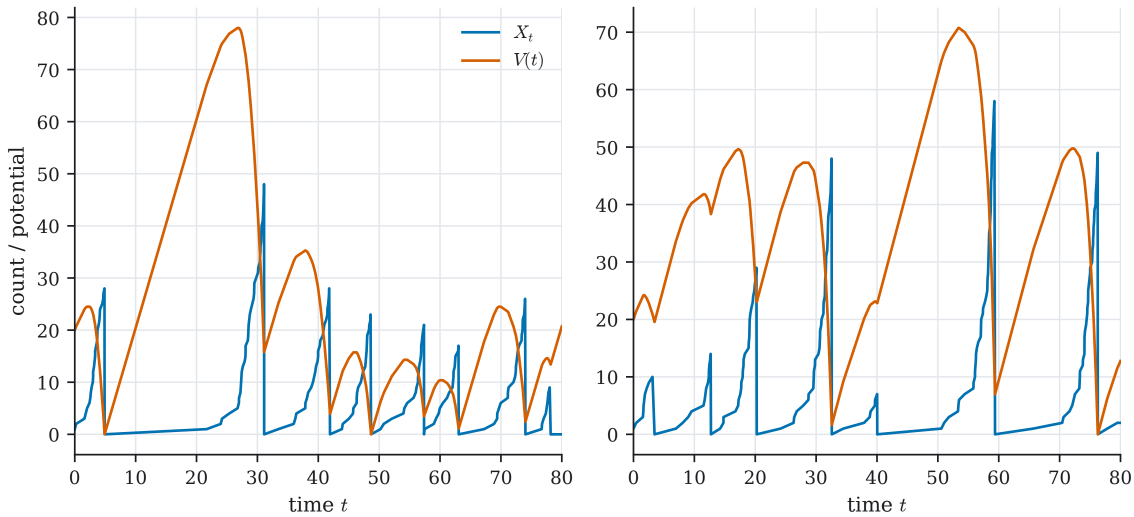


Figure 1.8: Two realisations of (1.26) and (1.27): potential $V(t)$ in vermillion, agency count X_t in blue ($\phi = 4$, $c = 0.8$, $\lambda = 0.5$, $\delta = 0.25$, $\omega = 0.15$). Every cycle has the same four parts in the same order — potential builds while X_t is small, X_t grows into the surplus, the surplus is consumed, the collapse follows — and no two are copies of one another.

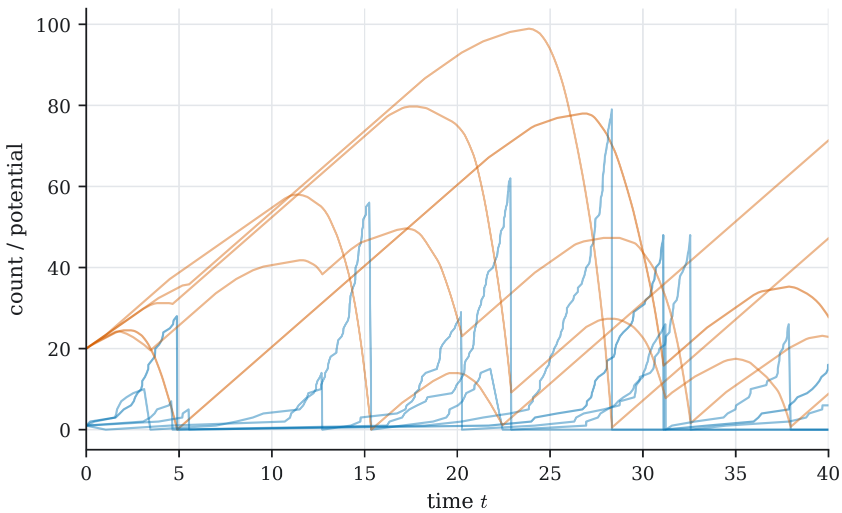


Figure 1.9: Several realisations from identical parameters and identical initial conditions. The first cycle is nearly common to all of them; later cycles are not, each catastrophe having reset the phase by a random amount. The parameters fix the amplitude, not the timing.

What X_t counts is deliberately left open. It may be individuals in a population, species within a genus, or autonomous units within an organisation, and the model does not distinguish them:

it is a proxy for order, or for locally reduced entropy, whose units are whatever the application supplies. That generality is also the model's principal weakness. Section 1.5 is the same hazard committed to one reading of X: there the units are types within a species, the potential is made by the population rather than supplied to it, and the catastrophe divides instead of destroying.

Open questions.

1. **The cycle period.** The distribution of the interval between successive catastrophes is a first-passage problem for the pair $(X_t, V(t))$, and is not known even in its mean. The figures establish that the period has substantial spread; how much, and how it depends on ϕ, c, λ and δ, is the quantity to go after.
2. **A quasi-stationary law.** Conditional on no catastrophe having occurred, does X_t have a quasi-stationary law? For the constant-rate birth–death–catastrophe process the answer is known and geometric (**????**), and the general theory of extinction times for catastrophe processes is available [16, 17]. Whether the state-dependent rate $\delta X_t/V(t)$ preserves that structure is the natural first question to ask of this model, and the one most likely to have an answer.
3. **What it could be fitted to.** The model has never been fitted to anything, and what data would identify ϕ, c and δ separately is not obvious. A time series of X_t alone almost certainly would not, V being unobserved and entering only through the ratio; the question is what second observable would suffice.

1.5 Speciation as catastrophe

1.5.1 The same hazard, with the inflow made endogenous

Section 1.4 drove catastrophe by a potential replenished from outside the population. This section keeps that hazard exactly and changes where the potential comes from: it is now produced by the population itself, in proportion to the ground it holds. Three consequences follow, and they are the reason the model is worth the space.

The first is that the cycle stops being a cycle. When inflow rises with area and cost rises with internal diversity, and nothing ever lowers the diversity but competition, a population is driven onto the locus where the two cancel and parks there. It does not oscillate; it sits at a marginal state and waits for the hazard.

The second is that the catastrophe divides rather than destroys. The units whose multiplicity drives the hazard are not individuals but *types*, and when the hazard fires they are not lost — they are promoted, all at once, to independence. Nothing dies. What changes is the law by which the survivors interact.

The third follows from the second. Because each catastrophe leaves behind products that obey the same law, the cycle branches instead of repeating, and no revival term is needed: at the settings used here a single founder gives rise to some 125,000 species by $t = 3000$, of which about fifty are then alive. Section 1.4 needed ω to keep 0 from absorbing; here the catastrophe restarts the process itself.

Set against section 1.2, the model is the other answer to the same question. There, speciation is an elementary birth at rate $\beta(t)$, the integrated rate converges, and S_∞ has a proper limit law. Here it is a catastrophe at rate $\delta X_i/V_i$, nothing converges, and the species count is the statistics of a branching cascade. The two are not rival descriptions of one phenomenon so much as the two ends of what a speciation rate can be made to mean.

The mechanism that charges the cost per type is what section 1.3.6 proposed and never ran. There, fitness carries a penalty $-\gamma S_t^{(a)}$ proportional to the species a population has accumulated

(1.25); here the potential carries a penalty $-cX_i$ proportional to the sub-species, and the model has been simulated. What is missing here is the competitive reversal, and what is missing there is a run.

1.5.2 State

The lattice is $L \times L$, with $L = 256$ throughout, so L^2 cells; the neighbourhood is Moore and the boundary periodic, so the lattice is a torus with no edge and no corner. Each cell is empty or occupied.

Every occupied cell carries two nested labels, a **species** and, within it, a **sub-species**. Only the sub-species is stored, the species being recovered from it, so that when a species fragments its sub-species are re-pointed at their new parents and every cell follows without being touched. For each extant species i the model carries

V_i	accumulated potential, a real variable obeying (1.31)
N_i	cells occupied by species i
X_i	number of *extant* sub-species of species i

with $X_i \leq N_i$, equality holding when every cell is its own sub-species. Both N_i and X_i are read off the lattice.

That X_i counts what is alive rather than what has ever existed is not a bookkeeping detail. A sub-species driven extinct by competition inside its parent stops costing that parent anything, so the coarsening of section 1.5.3 is a genuine regulator of dissipation and a species can hold its diversity, and hence its potential, in balance. Under a cumulative count the cost could only ever rise, and fragmentation would become certain from the moment a species began to diversify.

The run starts from one occupied cell at the centre, of one species with one sub-species, at potential V_0. Time is counted in **sweeps**: one elementary contest advances the clock by L^{-2}, so a sweep is one attempt per site on average. Competition is a discrete sequence of events while mutation and the potential are continuous, and the two are separated by advancing in chunks of a few hundred to a few thousand contests and applying the continuous updates once over each. That splitting is the model's only approximation in time, and its error is bounded by the movement of V_i within one chunk — a few parts in a thousand at the settings used here.

1.5.3 Competition

At each iteration a cell is drawn uniformly from all L^2 and one of its eight neighbours uniformly from the eight. If the two carry the same label, or both are empty, nothing happens. Otherwise they compete, and the winner's label is copied into the loser's cell.

Contests between occupied cells are one rule. Give each occupied cell ζ, carrying sub-species u of species i, the strength

$$w(\zeta) = \underbrace{V_i}_{\text{potential}} \times \underbrace{(n_\zeta + 1)^\theta}_{\text{local support}} \times \underbrace{\mathrm{e}^{-\nu|u|/\kappa_i}}_{\text{rare-type discount}}, \qquad \kappa_i = \frac{N_i}{X_i}, \tag{1.28}$$

where $n_\zeta \in \{0, \ldots, 8\}$ counts the neighbours matching ζ at the level of the contest, $|u|$ is the number of cells sub-species u occupies, and κ_i is the mean sub-species area within the parent. Let the contest go to ζ with probability $w(\zeta)/(w(\zeta) + w(\zeta'))$. Two readings follow.

- **Against a sub-species of the same species.** The parent's potential cancels from numerator and denominator, as it must, since two sub-species share one store. What survives is support and rarity, and n_ζ counts neighbours matching at the sub-species level.

- **Against a distinct species.** The rarity factor is omitted and n_ζ counts neighbours of the same *species*, so that for species A and B

$$\mathbb{P}(A \text{ wins}) = \frac{V_A(n_A+1)^\theta}{V_A(n_A+1)^\theta + V_B(n_B+1)^\theta}. \tag{1.29}$$

  This is the sense in which V_i is a potential: it buys territory, and it buys it only against other species.

Rarity is withheld between species deliberately. Inside a species it preserves the diversity that drives the model; between species it would merely blunt V_i, which is the quantity the model is about.

Contests against empty ground stand outside (1.28) entirely. Empty ground is a rival of fixed strength d, so an occupied cell takes a contested empty neighbour with probability $1-d$ and otherwise nothing happens — the occupied cell is never replaced, and empty ground never invades. Colonisation carries no dependence on V_i, which is a deliberate restriction: a species cannot widen its frontier by being rich, only take ground already held.

The support exponent θ buys surface tension. The factor $(n_\zeta+1)^\theta$ ranges over $[1, 9^\theta]$, so at $\theta = 3$ a cell with all eight neighbours of its own kind beats an isolated intruder better than 998 times in 1000: a domain pays along its perimeter and nothing in its interior, so it rounds up and can be eaten only from the edge. Without it boundaries wander and domains dissolve into filigree (figure 1.10).

The rarity exponent ν decides whether a burst is a catastrophe or a renaming, which is a sharper role than it looks. Since $|u|$ enters negatively, the *larger* sub-species is penalised, and at $\nu = 1.14$ one exceeding its rival by a mean sub-species area wins only about a quarter of their contests. What this changes is not how many sub-species a species carries but how large any one may grow. Set $\nu = 0$ and coarsening inside a species does what a voter model does — one sub-species engulfs the domain, and of the 7262 fragments the largest species would yield at $t = 2000$ exactly one is big enough to survive. At $\nu = 1.14$ the domain is instead a mosaic of comparable patches, and about half of 1534 fragments clear the threshold. Diversity in the bare sense survives either way: X_i climbs to the balance point under both. Only under the advantage does fragmentation actually multiply the species.

1.5.4 Mutation and the potential

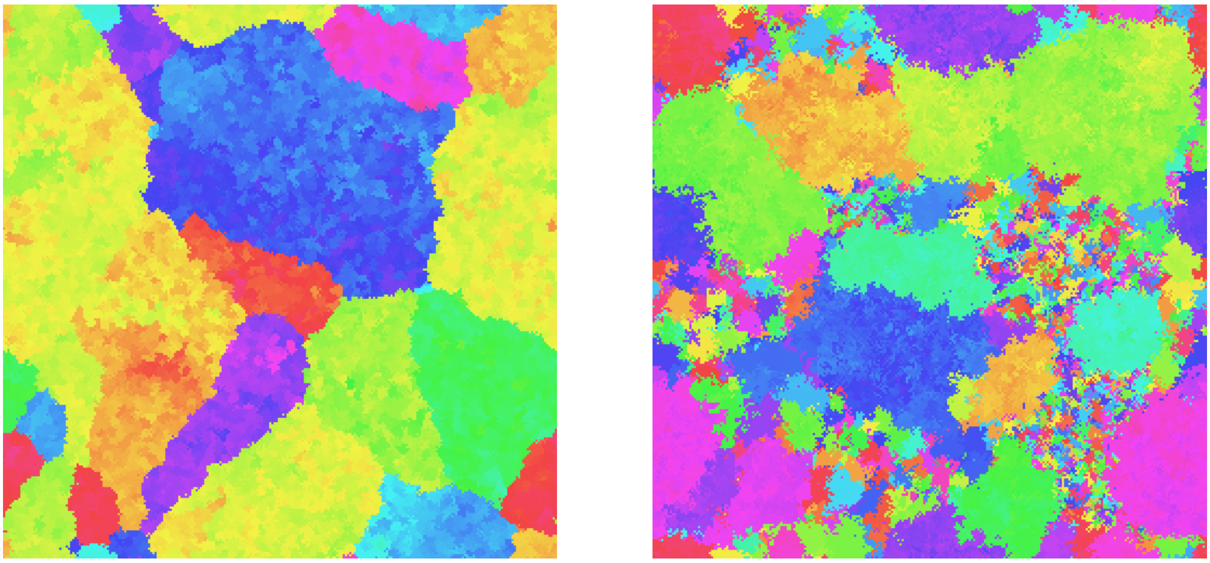
A sub-species arises when one cell mutates inside its species' domain, keeping its species label and taking a new sub-species label, so that $X_i \mapsto X_i + 1$; thereafter it spreads by the within-species contest. The rate is mV_i per cell, giving a species-wide rate

$$R_i^{\text{mut}} = m\, V_i N_i, \tag{1.30}$$

which couples novelty to prosperity: rich species diversify faster and thereby hasten their own dissipation, which is the feedback the model exists to display. The uncoupled alternative mN_i is available and changes the model's character, not merely its rate. One consequence deserves note: if the mutating cell was the last of its sub-species, that sub-species dies as the new one is born and X_i is unchanged. Mutation adds diversity only where there is something to mutate away from.

Each occupied cell supplies potential at rate φ and each extant sub-species costs c — the cost of multiplicity, charged per type rather than per head. Hence

$$\frac{\mathrm{d}V_i}{\mathrm{d}t} = \varphi N_i - c\, X_i, \tag{1.31}$$



support on, $\theta = 3$ — support off, $\theta = 0$

Figure 1.10: The same seed at the same time ($t = 2000$), with neighbour support on and off. With it, domains are compact and their boundaries short: seven per cent of occupied cells lie on a species boundary. Without it, boundaries wander, domains interpenetrate, and the boundary fraction rises to thirty-six per cent. The ragged state also carries far more species at once — 1335 against 26 — because nothing gathers a domain together and small holdings are never squeezed out. Hue labels the taxonomy and nothing else: it is carried along by species the competition rules have already made, and no cell ever changes hands because of a colour.

floored at zero, against section 1.4's $\mathrm{d}V/\mathrm{d}t = \phi - cX_t$. The two differences are the whole model: the inflow is charged per cell rather than supplied from outside, and the outflow is charged per type rather than per individual.

Three readings of (1.31) are worth recording.

(i) V_i does not appear on the right. It is a pure accumulator with no fixed point of its own, growing while $X_i < \varphi N_i/c$ and falling once diversity passes that threshold.

(ii) Since $X_i \leq N_i$, the most negative attainable drift is $N_i(\varphi - c)$, attained at $X_i = N_i$. A species can therefore lose potential at all only if $c > \varphi$, and that is the condition under which the catastrophe of section 1.5.5 is reachable. Below it there is no model: nothing ever bursts, and the run reduces to the lattice filling and then relabelling indefinitely.

(iii) A species carrying a single sub-species needs at least c/φ cells to break even — about nine at the settings used here. Below that area its potential drains away whatever it does, which is what governs the fate of the smallest fragments of a burst.

A species at $V_i = 0$ is not dead. It loses every between-species contest it enters against anyone with potential to spend, and stops mutating, but it still takes empty ground, and if its area exceeds cX_i/φ its potential recovers. It is merely defenceless until it does.

1.5.5 Fragmentation

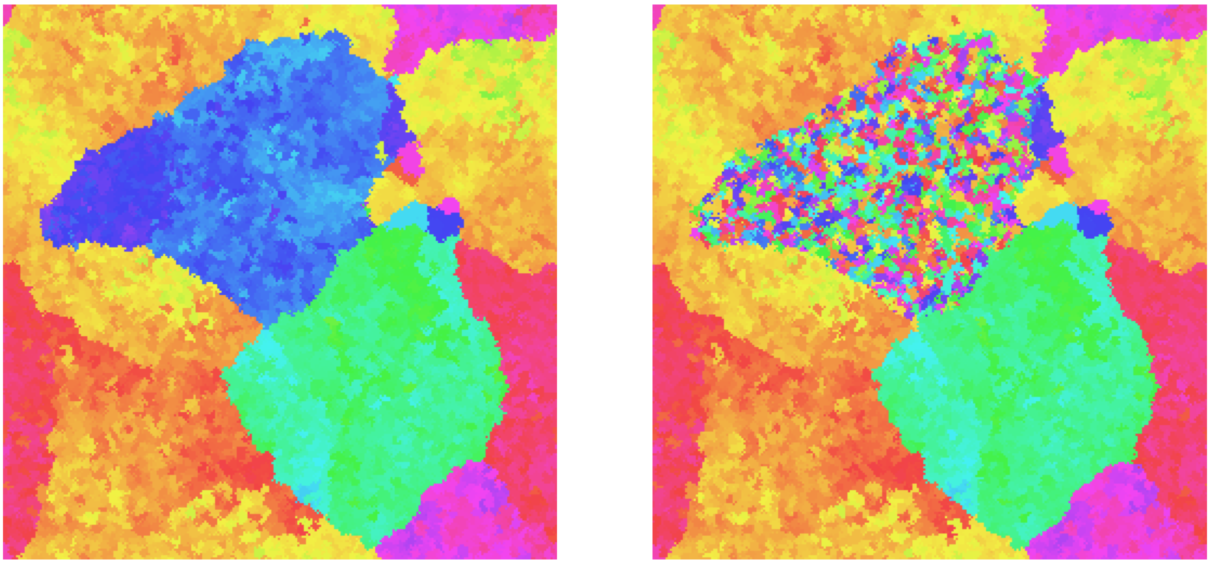
When a species has come to carry too many sub-species it **fragments**: every sub-species breaks away at once and becomes an independent species. The domains do not move. What changes is the law by which they interact, which switches from the within-species contest to (1.29): neighbouring patches that a moment earlier were contesting a shared inheritance now contest each other in proportion to their separate potentials.

Two triggers are live at once and tested in order. A species fragments deterministically if the update to (1.31) would take V_i to zero or below; failing that it fragments with hazard

$$R_i^{\mathrm{frag}} = \delta \, \frac{X_i}{V_i}, \tag{1.32}$$

which is (1.27)'s catastrophe rate $\delta X_t / V(t)$ with the count of individuals replaced by the count of types. Because (1.32) diverges as $V_i \downarrow 0$ it anticipates the deterministic trigger rather than competing with it, while also permitting early bursts in a species that is merely strained. At the settings used here the hazard does essentially all the work: a species in the regime of section 1.5.6 holds a potential of several hundred and never approaches zero, so it bursts because (1.32) fires and not because it is bankrupt. A species carrying one sub-species cannot fragment, there being nothing for a division to mean; it sits at $V_i = 0$ until its area recovers or it is overrun.

Each new species takes exactly the cells its sub-species held and begins with $X = 1$, its own only sub-species. Its potential is the one thing the burst does not settle by itself, the parent's store being wound up rather than divided by any natural rule, and it is reset to V_0. Two splitting rules were also implemented — equal shares, and shares proportional to area — but both interact awkwardly with the deterministic trigger, which fires precisely when there is nothing left to divide and so hands every fragment a potential of zero. Resetting is the only rule that behaves identically under both triggers.



before

after

Figure 1.11: A burst, across the single frame in which it happens, at $t \approx 1239$. The species held 12,826 cells — a fifth of the lattice — and still had $V_i = 755$ in hand when (1.32) fired; all 1444 of its sub-species became species at once, and it was the only species to fragment in that frame. Nothing has moved: the outline of the parent is exactly preserved and its neighbours are untouched. What has changed is the law inside that outline. Each fragment is drawn in a fresh hue here so that the division is unmistakable.

1.5.6 What the simulations show

Figures 1.12 and 1.13 are exact realisations at the settings above. Four consequences follow from the rules as stated.

The occupied region never retreats. No rule vacates a cell: contests against empty ground cannot, by construction, and contests between occupied cells only relabel. The occupied set is monotonically non-decreasing, and the lattice fills — here by about $t = 1100$. After that the dynamics reduce to relabelling, a biased voter model on a full lattice punctuated by mutation and fragmentation.

The population sits at the balance point. The cycle does not run away. Because mutation raises X_i while nothing lowers φN_i, a species is driven towards

$$X_i \approx \frac{\varphi N_i}{c}, \tag{1.33}$$

at which production and dissipation cancel and V_i goes flat. Figure 1.13 is that statement in one picture: the founder's potential climbs steeply while diversity is still cheap, flattens near 730 once X_i reaches (1.33) at about $t = 175$, and stays there. Every substantial species in a mature run is found on that locus to within a few per cent, with its potential parked at a few hundred and its drift near zero. The system settles into a marginal state rather than an oscillation, and what ends a species is not bankruptcy but the accumulating hazard of (1.32), which grows with the diversity the balance point obliges it to carry.

Fragments are filtered by area. A burst is not a fair division. Each fragment inherits its sub-species' cells, and those areas are broadly distributed; but by reading (iii) of section 1.5.4 a fragment of fewer than c/φ cells cannot pay for even its own single sub-species. Its potential drains within a fraction of a sweep, after which it loses every contest it enters and is overrun. Immediately after a large burst, hundreds of species may sit at $V_i = 0$ at once. The great majority of what a burst creates is gone almost immediately, and only the larger fragments survive to grow, diversify and burst in their turn — which is why a run that has produced tens of thousands of species holds only tens alive.

Growth and fragmentation overlap. At this colonisation rate the first turn of the cycle completes long before the lattice is full: the founder of figure 1.13 bursts on 8689 cells, a seventh of the ground, with open country still ahead of it. A species reaches the diversity that destroys it before it reaches the walls, so the run is never a sequence of clean generations.

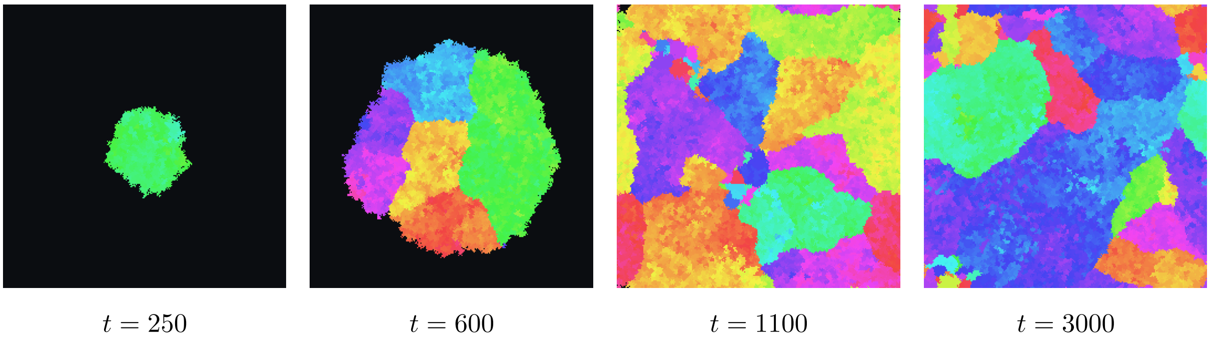


Figure 1.12: One run on a 256^2 lattice. At $t = 250$ a single species holds six per cent of the ground and already carries 443 sub-species, visible as mottling within one narrow band of hue. By $t = 600$ it has burst six times and the fragments are still spreading into open ground. The lattice closes at about $t = 1100$; by $t = 3000$ the run holds some fifty living species out of 125,000 ever created.

Open questions.

1. **Whether the balance point restores tractability.** This is the sharpest question the model raises. On (1.33) the potential is flat, so V_i is slowly varying at some $\bar{V}$, and the

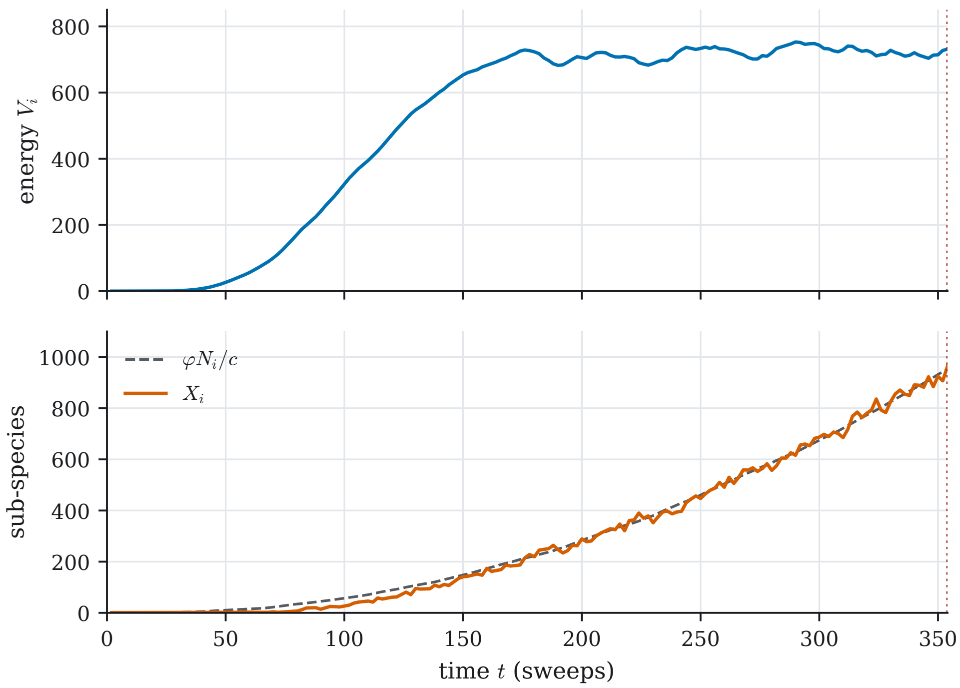


Figure 1.13: The founder species, followed from its first cell to the burst that ends it at $t = 354$. Potential V_i above, sub-species count X_i below against the balance point $\varphi N_i/c$ of (1.33); the dotted line marks the burst. The lower panel is the argument of section 1.5.6 in one picture — the dashed curve is area, rescaled, and diversity tracks it. The species is not bankrupt when it bursts: it still holds $V_i \approx 730$ on 8689 cells, but by then it carries 964 sub-species and the hazard (1.32) has grown large enough to fire.

hazard (1.32) becomes $\delta X_i/V_i \approx (\delta\varphi/c\bar{V})N_i$ — proportional to the species' *area*. But a catastrophe rate proportional to size is exactly the chapter on the birth–death–catastrophe process's δX_t. Whether the species count can therefore be read as a birth–death–catastrophe process with burst size X_i, with an effective δ of $\delta\varphi/c\bar{V}$, is the question to go after: it would put the chapter's least tractable model back onto the chapter's own machinery, and it is a conjecture supported by figure 1.13 rather than anything derived.

2. **The statistics of the cascade.** Section 1.4's open question was the distribution of the interval between catastrophes. Here the catastrophes branch, so the analogous object is the distribution of the number of *viable* fragments per burst — those clearing the c/φ threshold — and whether its mean sits above or below one decides whether the species count grows or collapses. The run holds tens alive out of 125,000 created, which suggests a mean near one and a process near criticality, but that has not been measured.

3. **A continuous species concept.** The model could be defined by intervals of hue rather than by discrete labels, with species identity a matter of proximity on the colour circle and speciation a matter of a lineage drifting far enough from its relatives to fall outside the interval. That would remove the taxonomy as a primitive and make it an observable, and it is not obvious whether the balance point survives the change.

4. **What it could be fitted to.** As with section 1.4, the model has never been fitted, and φ, c and δ enter the hazard only through the combination $\delta\varphi/c$ once the balance point is reached. A time series of the species count would not separate them; what second observable would is not known.

1.A The birth–plague model

Where section 1.4 drives the loss rate by a potential carried alongside a single count, this model drives it by a second count, so that what accelerates the loss of one component is the size of the other. It is recorded here as a specification; it has been simulated but not analysed.

Let X_t count uninfected individuals and Y_t the infected. In a vanishing interval,

$$\begin{aligned}\mathbb{P}(\Delta X_t = 1,\ \Delta Y_t = 0 \mid X_t, Y_t) &= \lambda X_t\,\Delta t + o(\Delta t),\\ \mathbb{P}(\Delta X_t = -1,\ \Delta Y_t = 1 \mid X_t, Y_t) &= (\delta + \chi Y_t) X_t\,\Delta t + o(\Delta t),\\ \mathbb{P}(\Delta X_t = 0,\ \Delta Y_t = -1 \mid X_t, Y_t) &= \mu Y_t\,\Delta t + o(\Delta t),\\ \mathbb{P}(\Delta X_t = \Delta Y_t = 0 \mid X_t, Y_t) &= 1 - \big[(\lambda + \delta + \chi Y_t) X_t + \mu Y_t\big]\Delta t + o(\Delta t),\end{aligned} \tag{1.34}$$

with λ the birth rate, δ the rate of spontaneous corruption, χ the contact-infection coefficient and μ the clearance rate of infected individuals.

The middle channel is written once and carries two mechanisms superposed. An uninfected individual is lost to corruption spontaneously at the constant per-capita rate δ, and to infection by contact at rate χ per uninfected–infected pair. Both produce the same jump, so the only thing that distinguishes them is how their rates scale: the first is linear in X_t, the second bilinear in $X_t Y_t$. Only that channel carries Y_t, but the coupling is two-way, since every infection removes one X and adds one Y.

The vitality-coupled variant. Carry a vitality $V(t)$ alongside the pair, replenished at a constant rate and drawn down in proportion to the population,

$$\frac{\mathrm{d}V}{\mathrm{d}t} = \phi - c\,X_t, \tag{1.35}$$

and divide the spontaneous part of the loss channel by it:

$$\mathbb{P}(\Delta X_t = -1,\ \Delta Y_t = 1 \mid X_t, Y_t) = \left(\frac{\delta}{V(t)} + \chi Y_t\right) X_t\,\Delta t + o(\Delta t), \tag{1.36}$$

the other three channels being unchanged. The reason for wanting the variant is that (1.34) has no mechanism for a collapse: its infected count rises when X_t is large and falls when it is small, which damps rather than amplifies. Dividing corruption by a vitality the population itself depletes supplies the missing amplification, since a population which has grown large drives V down and so raises its own loss rate — the ingredient section 1.4 used to close its cycle. Whether the combination gives a sustained cycle or a single collapse followed by a low-level steady state is the question the variant exists to ask. It has not been simulated.

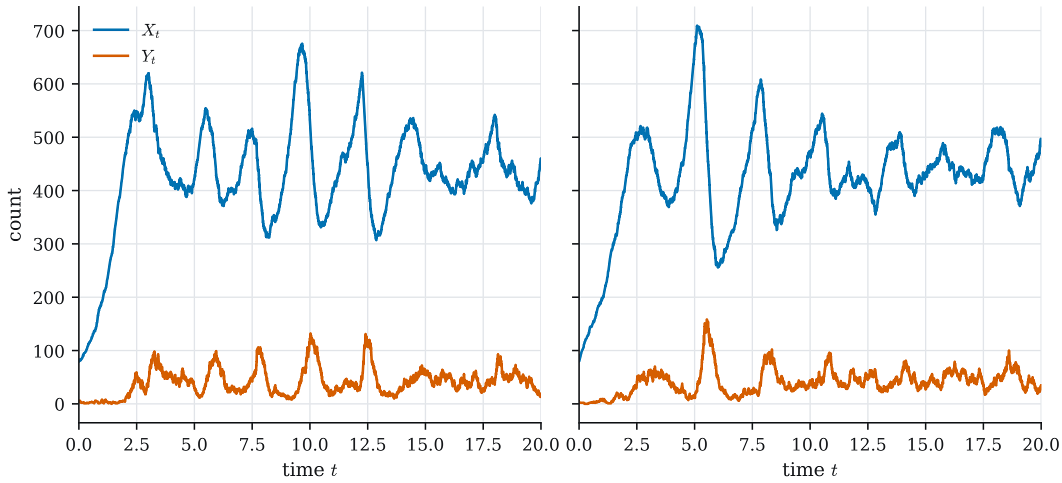


Figure 1.14: Two realisations of (1.34) at $(\lambda, \delta, \chi, \mu) = (1, 0.1, 0.02, 10)$: uninfected count X_t in blue, infected count Y_t in vermillion. The clearance rate is large relative to the contact rate, so infected individuals never accumulate and the epidemic acts as a brake on X_t rather than as a collapse of it. Set beside figure 1.9, the fluctuations never reset a phase.

1.B Public good accumulation

This model runs the chapter's feedback in the favourable direction: a larger population makes more of a good that raises its own birth rate. It is recorded here as a specification, together with the one quantity computed from it.

Let X_t count *Y. pestis* within one macrophage and $P(t)$ the concentration of public good in it. The good is produced in proportion to the bacterial count and cleared at a constant rate, and its presence raises the replication rate:

$$\begin{aligned}\frac{\mathrm{d}P}{\mathrm{d}t} &= g\,X_t - c,\\ \mathbb{P}(\Delta X_t = 1 \mid X_t) &= \lambda(t)\,X_t\,\Delta t + o(\Delta t), \qquad \lambda(t) = \lambda_0 + \alpha P(t),\\ \mathbb{P}(\Delta X_t = -1 \mid X_t) &= \mu\,X_t\,\Delta t + o(\Delta t),\end{aligned} \tag{1.37}$$

with g the production rate per bacterium, c the clearance rate, λ_0 the base birth rate, α the public-good advantage and μ the death rate, into which the effect of the host cell's toxic products is folded.

What stops the feedback being unconditional is the clearance term $-c$. While $X_t < c/g$ the good is cleared faster than it is made and P falls, so a small founding cohort may never start the loop at all. The regime worth attention is therefore $\mu > \lambda_0$, where the process is subcritical on its own and turns supercritical only once enough public good has accumulated, and where survival is not a matter of luck alone but of initial numbers.

The minimum founding cohort. Suppose the process embedded in a host-to-host sequence: if the infection within a macrophage survives to late time the cell dies and delivers its load into two more cells, and if not, the lineage ends. This is a Galton–Watson process, supercritical exactly when the probability of within-cell survival exceeds one half, and the quantity wanted is the smallest founding cohort achieving that:

$$K^* = \inf\left\{K :\ \lim_{t\to\infty} \mathbb{P}(X_t > 0 \mid X_0 = K) > \tfrac{1}{2}\right\}. \tag{1.38}$$

In the regime $\mu > \lambda_0$ the limit is zero for every K once the public good is removed, so K^* is finite only on account of it and measures exactly what the public good is worth: the number of bacteria that must arrive together for the mechanism to pay for itself. A general offspring law with mean m would replace the one-half by $1/m$, which moves K^* without changing what it measures.

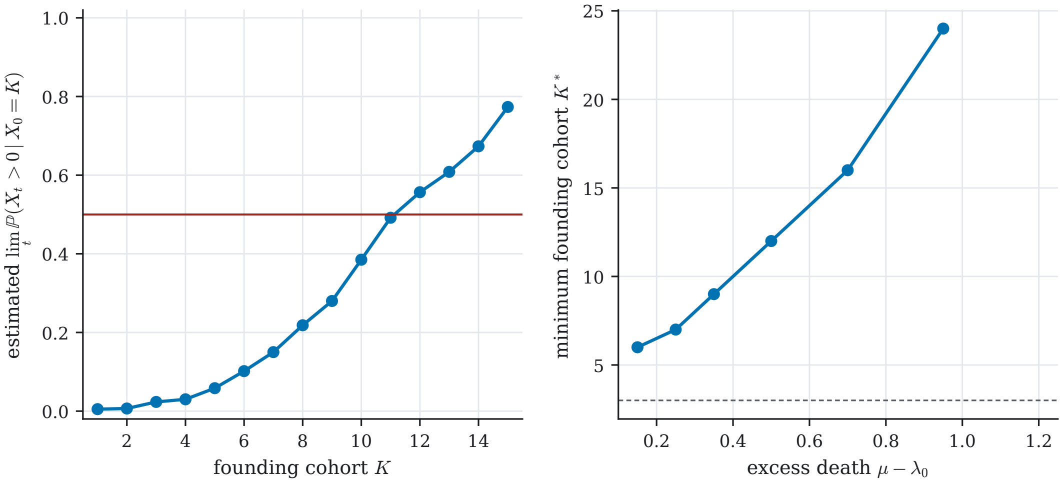


Figure 1.15: K^* computed at $(\lambda_0, \alpha, g, c) = (0.6, 0.25, 0.15, 0.45)$, so that P falls whenever $X_t < c/g = 3$, from 600 paths per point. The survival probability as a function of K is a steep sigmoid through one half, and K^* itself rises with the excess death rate $\mu - \lambda_0$. The dashed line in panel (b) is c/g: K^* sits above it, as it must, because a founding cohort smaller than the clearance threshold cannot start the loop. At $\mu - \lambda_0 = 1.2$ no $K \le 28$ crossed one half, which is a statement about the parameter point rather than a failure of the definition. This is one parameter slice and a numerical estimate, not a closed form: unlike **??**, which is a statement about a budget, K^* is defined by a limit of a survival probability and there is no reason to expect it in closed form.

1.C The inhomogeneous Yule generating function

The proof of proposition 1.2 is the standard generating-function computation for a pure birth process, with βt replaced by $\mathcal{B}(t)$. It is recorded here rather than in the body, because the method of characteristics is the mathematical introduction's and is not being rebuilt.

1.C.1 The generating function

The forward Kolmogorov equation for this system is that of the pure birth process with the rate carrying a t; writing $p_n(t) = \mathbb{P}(S_t = n)$ and expanding over a vanishing interval gives

$$\frac{\mathrm{d}p_n}{\mathrm{d}t} = \beta(t)(n-1)p_{n-1} - \beta(t)n\,p_n, \tag{1.39}$$

an instance of **??**. Multiplying by z^n and summing turns it into a first-order partial differential equation for $G(z,t) = \sum_{n\ge1} p_n(t)z^n$,

$$\frac{\partial G}{\partial t} = \beta(t)\,z(z-1)\,\frac{\partial G}{\partial z}, \tag{1.40}$$

and despite the time dependence this still yields to the method of characteristics, **??**. Introduce characteristic variables (s, τ) with

$$z(s, \tau = 0) = s, \qquad t(s, \tau = 0) = 0, \tag{1.41}$$

along which

$$\frac{\mathrm{d}G}{\mathrm{d}\tau} = 0, \qquad \frac{\mathrm{d}t}{\mathrm{d}\tau} = -1, \qquad \frac{\mathrm{d}z}{\mathrm{d}\tau} = \beta(t)\, z(z-1). \tag{1.42}$$

The first says G is a function of s alone and the second gives $t = -\tau$. With a single founding species $G(z,0) = z$, so on the initial curve $G = s$, and therefore $G = s$ everywhere: the whole problem is to express s in terms of z and t.

The third characteristic equation separates. Using $\mathrm{d}\tau = -\mathrm{d}t$ on the right,

$$\int \frac{\mathrm{d}z}{z(z-1)} = \int \beta(-\tau)\,\mathrm{d}\tau = -\int \beta(t)\,\mathrm{d}t, \tag{1.43}$$

and since the left side integrates to $\ln\big((z-1)/z\big)$ this is

$$\ln\left(\frac{z-1}{z}\right) = -\mathcal{B}(t) + \text{const.}$$

The initial conditions (1.41) fix the constant as $\ln\big((s-1)/s\big)$, because $\mathcal{B}(0) = 0$. Hence $(z-1)/z = \big((s-1)/s\big)\mathrm{e}^{-\mathcal{B}(t)}$, which rearranges to

$$s = \left(1 - \frac{z-1}{z}\,\mathrm{e}^{\mathcal{B}(t)}\right)^{-1},$$

and so

$$G(z,t) = \frac{z}{z + (1-z)\,\mathrm{e}^{\mathcal{B}(t)}}. \tag{1.44}$$

Differentiating with respect to z and setting $z = 1$ returns $\mathbb{E}(S_t) = \mathrm{e}^{\mathcal{B}(t)}$, which is (1.8) again by a different route.

1.C.2 The distribution

Write $p = \mathrm{e}^{-\mathcal{B}(t)}$ and divide numerator and denominator of (1.44) by $\mathrm{e}^{\mathcal{B}(t)}$:

$$G(z,t) = \frac{pz}{1-(1-p)z}, \tag{1.45}$$

which is the probability generating function of a geometric law on $\{1, 2, \dots\}$ with success parameter p. That is (1.7), and it completes the proof of proposition 1.2. Repeated differentiation of (1.44), which was the route originally taken, gives $\partial_z^n G = n!\,\mathrm{e}^{\mathcal{B}}(\mathrm{e}^{\mathcal{B}}-1)^{n-1}(z(1-\mathrm{e}^{\mathcal{B}})+\mathrm{e}^{\mathcal{B}})^{-(n+1)}$, and the inversion formula $\mathbb{P}(S_t = n) = (n!)^{-1}\partial_z^n G|_{z=0}$ of **??** returns the same law.

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